

Task 3: Basic Image Filter App

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Task Title

Create a Basic Image Filter App

Objective

The objective of this task is to understand the basics of image processing using Python and OpenCV. The application allows users to apply different filters to an image and save the processed output.

Tools and Technologies Used

- Python
- OpenCV (cv2)
- Google Colab
- GitHub

Introduction

Image processing is an important field of Artificial Intelligence and Computer Vision. It helps computers analyze and modify images to extract useful information or improve image quality. OpenCV is a popular Python library used for image processing tasks.

Procedure

Step 1: Upload an Image

An image named sample.jpg was uploaded to Google Colab for processing.

Step 2: Load the Image

The image was loaded using OpenCV's `cv2.imread()` function.

Step 3: Apply Grayscale Filter

The image was converted into grayscale format using:

```
cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

This filter removes color information and displays the image in shades of gray.

Step 4: Apply Blur Filter

A blur effect was applied using:

```
cv2.GaussianBlur(image, (15,15), 0)
```

This reduces image details and smooths the image.

Step 5: Save the Output

The processed images were saved using `cv2.imwrite()`.

Code Implementation

```
import cv2
from google.colab.patches import cv2_imshow

img = cv2.imread("sample.jpg")

choice = input("Enter filter (gray/blur): ")

if choice == "gray":
    output = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    cv2_imshow(output)
    cv2.imwrite("grayscale.jpg", output)

elif choice == "blur":
    output = cv2.GaussianBlur(img, (15,15), 0)
    cv2_imshow(output)
    cv2.imwrite("blur.jpg", output)

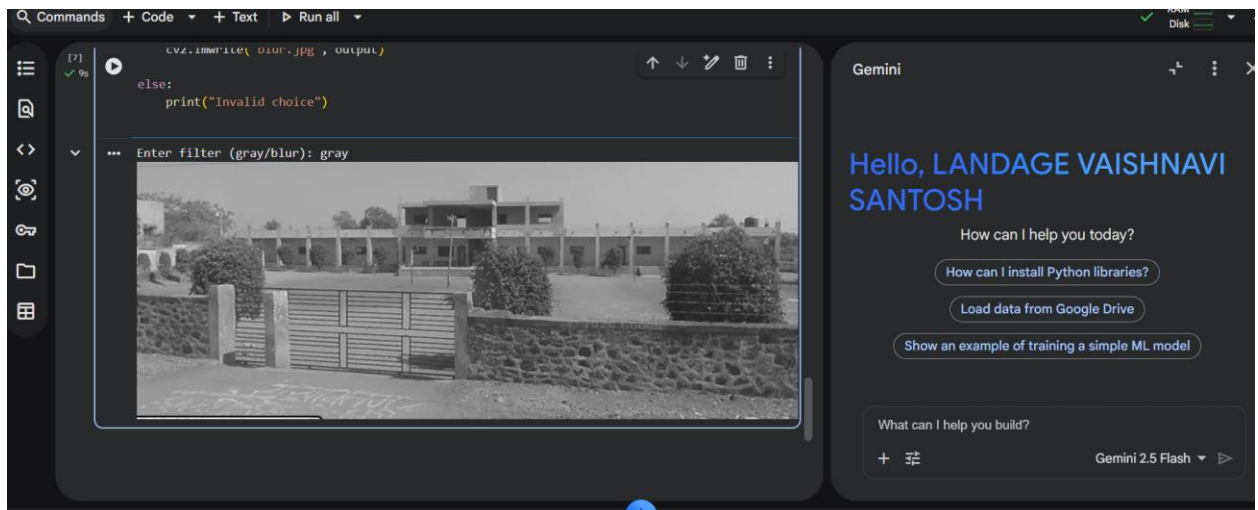
else:
    print("Invalid Choice")
```

Results

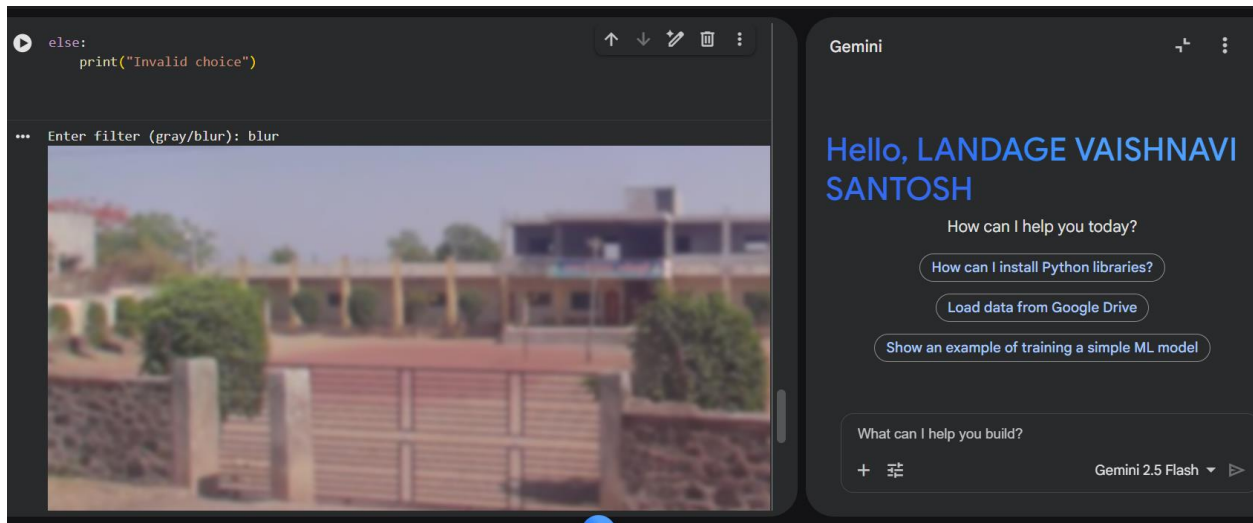
Original Image



Grayscale Output



Blur Output



Observations

- The grayscale filter removes color information from the image.
- The blur filter smooths the image and reduces sharp details.
- OpenCV provides simple functions for image manipulation.
- User input allows selection of different filters.

Learning Outcomes

- Learned how to load images using OpenCV.
- Learned how to apply grayscale and blur filters.
- Learned how to save processed images.
- Gained practical experience with image processing concepts.

Conclusion

This task successfully demonstrated the implementation of a basic image filter application using Python and OpenCV. The application allowed the user to apply grayscale and blur effects and save the resulting images. This task helped in understanding the fundamentals of image processing and computer vision.